

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems

Experiment: 9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR (dB).

Program:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

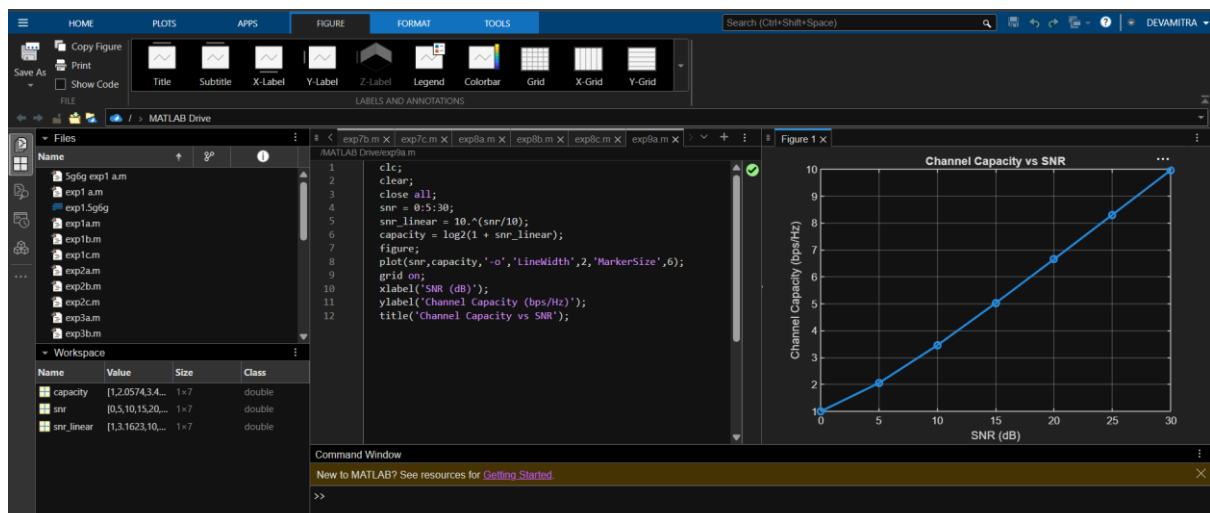
grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

Output:



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

Program:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

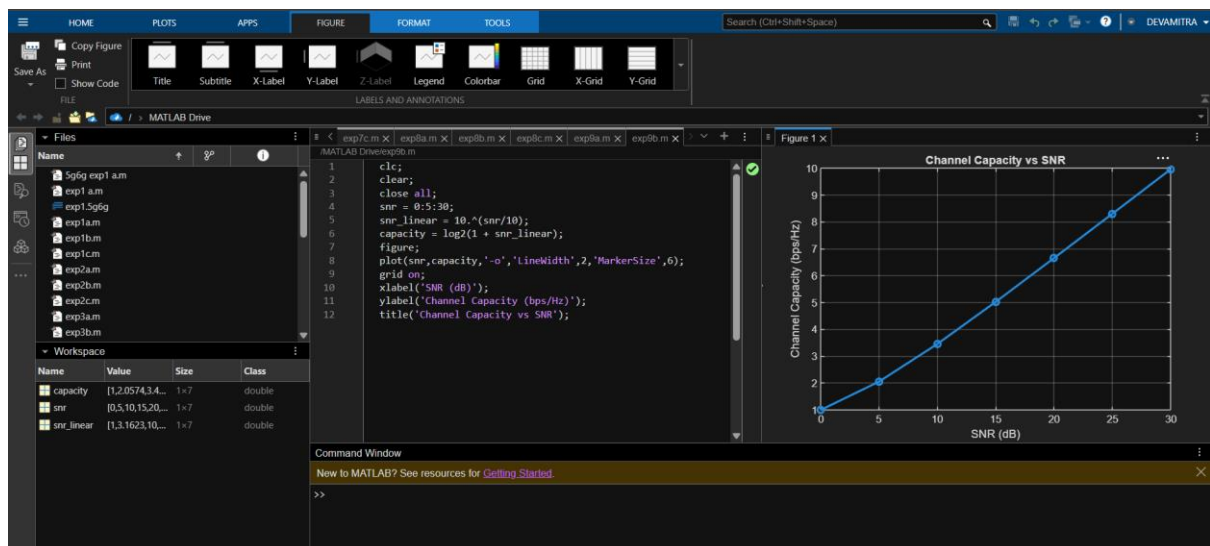
grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

Output:



Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

Program:

```
clc; clear; close all;
```

```
antennas = [4 8 16 32 64];
```

```
gain = 10*log10(antennas);
```

```
bar(antennas, gain);
```

```
grid on;
```

```
xlabel('Number of Antenna Elements');
```

```
ylabel('Antenna Gain (dB)');
```

```
title('Antenna Gain vs Number of Antennas');
```

Output:

